

Sleep-related Breathing Disorders

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Disorders of breathing related to sleep are relatively newly recognized and less than fully understood. This review presents the terminology used to describe them, and describes the physiology of sleep and of control of ventilation, the pathophysiology of breathing disorders during sleep, their various clinical manifestations, current diagnostic techniques, and the treatment modalities available at present. Among the diagnostic approaches discussed are airway fluoroscopy during sleep, pneumography, and polysomnography. Approaches to medical and surgical management of these disorders are reviewed.

Speculation regarding the underestimation of the prevalence of these disorders, the male predominance, and their relationship to snoring, coronary artery disease, and hypertension, which also show male predominance, are presented. Also suggested is a relationship of sleep apnea, obesity, and mental retardation in childhood-onset or congenital disorders such as Down's syndrome and Prader-Willi syndrome, and in other endocrine dysfunction diseases.

Over the past two decades awareness of a variety of respiratory difficulties related to sleep has been increasing. In 1956, Burwell et al.¹ described the dramatic pickwickian syndrome tetrad of obesity, alveolar hypoventilation, hypersomnolence and cor pulmonale. Further appreciation of a number of incomplete (pickwickian) syndrome variants,² respiratory control abnormalities,³⁻⁷ intoxications,⁸ and respiratory apparatus abnormalities⁹⁻¹⁴ have led to greater awareness of the prevalence and variability of the sleep apnea syndromes. In addition, it has become apparent recently that abnormalities of adult and childhood behavior (performance), hypertension, sexual dysfunction,¹⁵ sudden in-

fant death syndrome^{16, 17} and a variety of psychosocial disturbances may be related to disorders of breathing during sleep. It is now well documented that sleep apnea occurs in apparently normal patients. It should be pointed out that the prevalence of sleep apnea is not known, but that it is probably significantly underestimated because it frequently is relatively "asymptomatic." Also, in spite of our growing awareness, we continue to be underaware of this relatively new entity.

In order to understand the disorders of breathing during sleep one must know something about breathing, about sleep, and about their interactions. This review describes what is cur-

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rently known and discusses some of the information and speculation regarding disorders of breathing during sleep.

Information regarding diagnosis and management is also presented.

Definition of Terms

Sleep disorders are described by a number of terms, often misunderstood and misused. To avoid confusion, some of the pertinent terms are defined here:

Apnea is the cessation of airflow at the levels of the nares and mouth. The durations of pathologic apneic episodes vary with age. In adults the duration is 10 seconds.¹⁵

Periodic breathing is defined as at least two central apneic events lasting less than 10 seconds within 20 seconds of each other.¹⁸

Age-specific standards for infants were outlined in a recent review edited by Guilleminault and Korobkin¹⁸: "No full-term control infant has been recorded with 'cessation of air exchange' during sleep longer than 15 seconds between 1 and 6 months of age. Apnea during sleep longer than 15 seconds appears to be pathological up to 6 months after birth. For premature infants prior to 40 weeks' gestational age, the duration of normal apnea is extended to 20 seconds. Apneas during sleep longer than 10 seconds in full-term controls within the same age range (1 to 6 months) were also considered 'rare to exceptional' during hospital surveillance and should lead to careful evaluation."

Central apnea is the cessation of both airflow and respiratory movements.

Obstructive apnea is the cessation of nasal and oral airflow with persistence of respiratory efforts.

Mixed apnea is the cessation of airflow and an absence of respiratory efforts initially, with subsequent resumption of efforts and finally, reestablishment of airflow.

Sleep apnea syndrome is diagnosed with the observation of at least 30 apneic episodes in both rapid eye movement (REM) and non-REM sleep, some of which appear repeatedly in non-REM sleep, during seven hours of unsedated nocturnal sleep.¹⁵

REGULATION OF RESPIRATION

This section deals with a review of the basic physiology of respiratory control, subdivided into chemoreflexes and mechanoreflexes.

Chemoreflexes

Central (Medullary, MCR) Chemoreceptors

Medullary chemoreceptors are located superficially on the ventral medullary surface of the brainstem bilaterally. These receptors are responsive to changes of the partial pressure of carbon dioxide (PCO_2) or of the level of hydrogen ion (H^+) in the local extracellular fluid. Molecules of carbon dioxide (CO_2) diffuse freely across the blood-brain barrier, so that the PCO_2 of arterial blood (the $[H^+]$ of the cerebrospinal fluid as it varies with the PCO_2) is the main determinant (acutely) of medullary chemoreceptor activity. The pH of arterial blood is not an acute determinant (because of relative impermeability of the blood-brain barrier to H^+). Upon stimulation of the medullary chemoreceptors by an increase in PCO_2 , ventilation is increased.

Peripheral (Carotid Body) Chemoreceptors

Carotid body chemoreceptors are located bilaterally at the level of the bifurcation of the common carotid artery. These receptors are responsive to arterial blood PCO_2 and H^+ , as well as to the partial pressure of oxygen (PO_2) in arterial blood. Stimulation by a rise in PCO_2 or decrease in pH or PO_2 results in increased neural activity, which is transmitted to brainstem respiratory centers via the carotid sinus nerve (branch of the glossopharyngeal nerve). Increased activity results in an increased level of ventilation. Hypoxia potentiates peripheral carotid body responsiveness to hypercarbia; hyperoxia diminishes the response.

Mechanoreflexes

These receptors consist of bronchopulmonary and chest wall subgroups. The bronchopulmonary receptors are of three types: pulmonary stretch receptors, lung irritant receptors and type J receptors. Impulses from these receptors travel to brainstem respiratory centers via cervical vagal fibers.

Pulmonary stretch receptors are situated in the walls of the airways and are innervated by myelinated vagal afferent fibers. Lung inflation provides the stimulus, with reflex changes being termination of inspiration (the Hering-Breuer reflex) and prolongation of expiration.

Lung irritant receptors are located within the

airway epithelial lining. Transmission centrally is also via myelinated vagal afferents. Reflex activities in response to various noxious and irritating agents consist of termination of inspiration and prolongation of expiration.

Type J receptors are thought to be located in the alveolocapillary interstitium with central projection via unmyelinated vagal fibers. Stimulation results when there is distortion of the interstitial space (e.g., venous congestion). The reflex response to stimulation is rapid, shallow, tachypneic breathing.

The significance of mechanoreflex control of normal ventilation is unclear, since the magnitude of stimulation required is nonphysiologic.

SLEEP

The physiologic study of sleep began in 1953, when Aserinsky and Kleitman²⁰ first described REM sleep and its associations with bursts of eye movements, autonomic changes during dreaming, and a characteristic electroencephalographic pattern. Early students of sleep noticed two distinct types of sleep behavior patterns, referred to as rapid eye movement (REM, active, paradoxical, desynchronized) sleep and non-rapid eye movement (non-REM, NREM, quiet) sleep.^{21, 22}

Adult humans fluctuate through four stages of non-REM sleep. There are fairly characteristic electroencephalographic changes associated with each, but for the purposes and scope of this review certain simplifications are used. Along the continuum through stages of wakefulness 1→2→3→4 of non-REM sleep, the waves of the electroencephalogram progressively decrease in frequency and increase in amplitude as sleep becomes deeper. These changes are associated with a decrease in generalized muscular activity. In contrast, REM sleep has been classically described by an electroencephalogram not unlike that obtained during wakefulness, associated with binocularly synchronous eye movements (the so-called rapid eye movements) and phasic bursts of generalized muscular activity. Paradoxically, in spite of the phasic muscular activity, submental, nuchal and postural muscle tone is generally decreased during REM sleep.²² REM sleep is further subdivided into tonic and phasic periods. The tonic (inactive) phase is characterized by electroencephalographic desynchronization and decreased general muscular tone. The

phasic (active) period, on the other hand, is characterized by bursts of rapid eye movements and muscular activity, as well as by a marked increase in the irregularity of respiration.

Rapid eye movement sleep is the stage associated with dreaming, and is also known to be associated with numerous autonomic changes involving respiration, circulation,²³ temperature regulation²⁴ and cerebral blood flow.²⁵

The exact purpose of sleep is unknown, and the interrelationships of the REM and non-REM portions in health are unclear. Likewise, the purposes of respiratory adaptations to sleep are poorly understood. Important and useful in the investigations involving sleep are the remarkable phylogenetic similarities of sleep patterns. Among various species, REM sleep and non-REM sleep (also known in animals as slow-wave sleep or SWS) are quite similar and comparable, as are the respiratory adaptations to sleep.^{26, 27}

VENTILATORY RESPONSES DURING SLEEP

The study of ventilation and chemoresponsiveness during sleep has become increasingly sophisticated and informative. Polysomnography is rapidly becoming standardized and accepted. This technique employs multiple physiologic monitors during sleep.

The general setting of such laboratories is a room resembling a bedroom. The patient arrives around his ordinary bedtime, and the equipment is connected and prepared. The measurements made are similar to those listed:

1. Oral and nasal air flow are determined by a thermistor or by a CO₂ analyzer to document the flow of air at the airway.
2. Abdominal movement and thoracic movement are assessed to document presence or absence of respiratory effort. These two systems allow differentiation between obstructive and central apnea.
3. Abbreviated electroencephalography (using the C₃/A₂ to C₄/A₁ leads of the international 10–20 system) is used for sleep staging.
4. Electro-oculography is used to ascertain the presence or absence of rapid eye movements, which assists in establishing the stage of sleep.
5. Electromyography is used on certain airway muscles (mental, submental groups) to document the levels of activity of these muscles and to correlate such activity with other parameters.
6. Electrocardiography is done.
7. Ear oximetry is used to document the degree of

desaturation that may occur and to correlate such desaturation with other physiologic events.

Simultaneous measurements by use of these modalities allow sleep staging, cardiorespiratory monitoring, and differentiation between upper airway obstruction and central apneic episodes.

Also utilized are the special studies of chemoreceptor responses, i.e., sleep-related ventilatory responses to manipulation of the fractional concentration of inspired carbon dioxide (FI_{CO_2}) and the fractional concentration of inspired oxygen (FI_{O_2}) to evaluate basic underlying abnormalities. Much of the investigative work done to date has been done in animals. The validity of extending the conclusions drawn from such studies to human responses depends largely upon the similarity of sleep behavior among different species. Such generalizations, although they appear to be reasonable, may not be altogether valid.

A rise in the P_{CO_2} in arterial blood (P_{aCO_2}) during sleep was first described by Endres,²⁸ in 1923. Subsequent observations clarified that the P_{aCO_2} is normally increased during non-REM sleep by an average of 3 to 8 mm. Hg.^{29, 30} Reports have confirmed that the increase is due primarily to a sleep-induced reduction in ventilatory drive. In dog studies, Phillipson and coworkers³¹ found that the P_{aCO_2} during REM sleep, compared with slow-wave sleep, was further increased. They also observed that the P_{aCO_2} at arousal points and the duration of rebreathing tolerated prior to arousal are much greater in REM sleep. In cats, Wurtz and O'Flaherty³² found the P_{aCO_2} to average 4 mm. Hg higher in REM sleep than in slow-wave sleep; however, Guazzi and Freis³³ found a difference of only 1 mm Hg.

In 1976, Fagenholz et al.³⁴ reported no difference in P_{aCO_2} in human infants in a comparison of REM with non-REM sleep states. It should be pointed out that changes occur very rapidly in REM sleep, and representative or average points for sampling are difficult to judge. The choice of phasic vs. tonic periods may prove to be the key to these conflicting observations.

As previously mentioned, REM sleep is strikingly nonhomogeneous. The breathing pattern during the phasic segments (associated with bursts of eye movement) is rapid, shallow, and more irregular than that associated with tonic periods. In adults, during periods characterized by bursts of increased muscular activity, alveolar ventilation may even be greater. During apneic

periods that occur in REM sleep, the arterial blood oxygen saturation does decrease.³⁵ Thus, P_{aO_2} may be low, high, or unchanged over an entire period of REM sleep.³³ With respect to responsiveness to changes in P_{aO_2} , the characteristic REM sleep breathing irregularity is not altered by isocapnic hypoxia. However, as arterial oxygen saturation decreases in REM sleep, an overall increase in ventilation occurs, indicating preservation of the hypoxic response.³¹ Hyperoxia during REM sleep causes a mild ventilatory depression, although the irregular pattern of breathing is not altered. During slow-wave sleep in dogs, alveolar ventilation and respiratory frequency are increased in response to decreasing arterial blood oxygen saturation, and the arterial blood oxygen saturation at arousal is very reproducible. Hypoxic arousal occurs at a lower arterial blood oxygen saturation in REM sleep than in slow-wave sleep.³¹

RESPIRATION DURING SLEEP IN SLEEP APNEA PATIENTS

Although normal adult patients show characteristic sleep-related changes in ventilatory patterns and have apneic periods during sleep, there is a clear-cut difference between the number of apneic events in the sleep of such patients compared with patients who have sleep apnea. In adult sleep apnea syndrome, the number of episodes of apnea is usually several hundred per night. The group at Stanford reports this as the Apnea Index (total number of 10-second apneic events divided by the total number of hours slept, or roughly, the number of occurrences of apnea per hour). Among patients diagnosed as having sleep apnea, the lowest Apnea Index was 11, compared with a maximum of 4 recorded for one member of the control group.¹⁵

DIAGNOSTIC TECHNIQUES

Clinical Aspects

Because of the relative lack of specific physical findings and the protean symptoms that we now realize may be part of these syndromes, the review of systems may indeed be most important in diagnosing such disorders. Based on the descriptions of the patient populations, adult and pediatric, studied by the Stanford group¹⁵ for

sleep apnea complaints, certain characteristics emerge as common. There is a very striking male predominance (20:1 to 60:1) in every report of sleep-apneic patients to date.^{15, 36-38} Adult patients most frequently seek medical advice because of daytime hypersomnolence. Approximately 80 per cent of adult patients who have sleep apnea involving upper airway disorders have this as their chief complaint.¹⁵ Other initial complaints include loud snoring (frequently described by the bed partner as associated with respiratory pauses and snorts), abnormal movements during sleep, morning headaches, depression, personality change, marital difficulties, deteriorating intellectual performance, and unexplained hypertension or heart failure. Patients reporting severe hypersomnolence often described the automatic behavior syndrome: retrograde amnesia for periods of minutes to several hours, during which simple, repetitive tasks are possible, but complex behavior is not, and hypnagogic hallucinations (appearance of images at the time of maximum sleepiness).

Children, on the other hand, while they may have hypersomnolence, are more likely to be brought to medical attention because of poor school performance, "behavior disorders," or relapse of enuresis in a previously completely toilet-trained child. The parents may have observed snoring or chest movements without apparent airflow during sleep. Children who have facial anomalies or histories of facial or airway surgery are frequently found to have sleep obstruction.

At first glance sleepiness may seem of minor importance. However, as the number of patients with sleep apnea syndromes increases, the impact of this symptom assumes greater proportions, and most patients perceive it as their greatest problem. Job losses and marital difficulties, debilitating depressions sometimes associated with suicide attempts, and automatic behavior are important manifestations of the narcolepsy-sleep apnea continuum. Many patients describe falling asleep during their favorite sports event, while driving, and during intercourse.

Snoring is generally considered a trivial nuisance. Recently, the relationship of snoring to obstructive (and possibly secondary central) apnea in sleep was investigated. Lugaresi³⁹ suggests that snoring is a part of the spectrum of hypersomnolence with periodic apnea. According to Lugaresi, snoring is a hypnogenic stenosis of the upper airway. He has demonstrated that heavy snorers have apneas indistinguishable

from those of patients with hypersomnolence with periodic apneas, and that pickwickian patients may have snoring alone following weight losses.

The clinical significance of the progression along this continuum is further supported by the observation that loud snorers may have alveolar hypoventilation and pulmonary and systemic arterial hypertension intermediate between the values of healthy persons and patients with hypersomnolence with periodic apnea. Lugaresi

TABLE 1. Differential Diagnosis of Sleep-related Breathing Disorders Classified by Site of Abnormality

Central control
Primary alveolar hypoventilation (Ondine's curse)
Bulbar disturbance secondary to polio, tumor, granulomatous process, infarction, trauma, surgery
Phrenic paralysis
? Prematurity, sudden infant death syndrome
Arnold-Chiari malformation
Dandy-Walker syndrome
Neurosurgical procedures
Secondary alveolar hypoventilation
Ethanol, narcotics, sedative hypnotics, anticonvulsants,
? chronic obstructive pulmonary disease
Cervical spine surgery
? Down's syndrome
Peripheral control
Mobius syndrome
Cervical cordotomy (bilateral)
Drugs
? Prematurity, sudden infant death syndrome
? Chronic obstructive pulmonary disease
? Down's syndrome
Muscular
Muscular dystrophies
Shy-Drager syndrome
? Prematurity, sudden infant death syndrome
Myotonic dystrophy
? Chronic obstructive pulmonary disease
Airway obstruction
Orofacial abnormalities (Pierre Robin syndrome, Treacher Collins syndrome, primary micrognathia, cleft palate)
Adenoidal and/or tonsillar hypertrophy
Obesity, pickwickian syndrome
Acromegaly
Tumors
Infection, nasopharyngitis
Postoperative edema
Hypothyroidism
Post-facial or airway surgery
Kyphoscoliosis
? Down's syndrome
Excessive relaxation of genioglossus and pharyngeal muscles
Combined or unclassified
Pickwickian syndrome
Sequelae of poliomyelitis and other acute neurologic processes such as Guillain-Barré syndrome
? Bird-like face syndrome (acquired micrognathia, hypersomnia and sleep apnea)
? Long-standing upper airway obstruction
? Chronic obstructive pulmonary disease

speculated that with time and worsening of airway configuration, either by age (tone) or by progressive obesity, a transition occurs. The snoring patient may first notice exaggerated diurnal somnolence, develop hypersomnolence with periodic apnea and central respiratory depression, then proceed to polycythemia, cor pulmonale, and even death.

Among the Stanford sleep-apneic patients, the prevalence of snoring is extremely high, and it is often reported as having been present for many years. Parents of pediatric sleep-apneic patients reported noticing snoring in children as young as a few months of age.¹⁵

Snoring, obesity, heart disease and hypertension show a 2 to 1 male to female predominance that correlates well with the striking male predominance in the disorders of breathing during sleep.

Physical examination of these patients often shows no abnormality. Approximately 5 per cent of sleep-apneic patients have weights compatible with the pickwickian physique, although a third can be considered overweight. Very few of the male patients reported had obvious head or neck abnormalities. However, among the few women, micrognathia was a much more frequent finding. Approximately 60 per cent of all patients have been clearly hypertensive.¹⁵ Important supportive evidence for a sleep apnea syndrome is provided by the presence of hypertension, a loud pulmonary component of the second heart sound, digital clubbing, nasal polyps, obstructed nasal airways, an obstructed oral airway, stridor, facial, nuchal or thoracic anomalies, a habitus of extreme obesity, or abnormal endocrine function.

Tables 1 to 4 describe the differential diagnosis and presenting age-related attributes of patients with the sleep apnea syndromes.

Roentgenographic Methods

Routine radiographic examinations of the chest and airway are generally of little value in this spectrum of disorders. Such studies are

TABLE 2. Attributes of Infants at Risk for Sleep-related Breathing Disorders

Previous episode of "near miss"
History of a sibling with sudden infant death syndrome or "near miss"
History of prematurity
History of periodic breathing
History of adult relative with a sleep/breathing disorder
Facial anomaly
Surgery for facial or airway anomaly

TABLE 3. Attributes of Children at Risk for Sleep-related Breathing Disorders

? History of "near miss" as an infant
History of noisy or snoring respirations during sleep
History of nasal (allergic) airway obstruction
? Sibling with sudden infant death syndrome, "near miss SIDS" or obstructive sleep
History of surgery for airway repair (cleft palate, velopharyngeal insufficiency)
Abnormal postures during sleep
Sleepwalking
Unexplained hypertension
Severe obesity
Progressive behavioral abnormalities, personality change, deteriorating school performance
? Medication for seizure disorder (phenobarbital) or cough suppression
Daytime sleepiness
Excessive appetite
Recurrence of enuresis

usually performed on awake subjects. Recent reports have described the use of airway fluoroscopy and cineradiography during sleep to make the specific diagnosis.^{40, 41} The technique involves allowing the patient to fall asleep on the fluoroscopic table and examining the airway during sleep. Under ideal circumstances, the study is repeated with the patient awake to document continual airway patency. Such studies of non-sleep apneic subjects have shown no obstruction of the airway during sleep.⁴⁰ This technique is noninvasive, and the only risk is that of the radiation exposure. In the investigation of sleep-breathing disorders, non-invasiveness has more than aesthetic appeal, for airway violation for monitoring purposes may interfere with detection of the abnormality.

TABLE 4. Attributes of Adults at Risk for Sleep-related Breathing Disorders

History of snoring
Complaints of bed partners—abnormal snorting, respiratory pauses, abnormal motor activity
Excessive sleepiness during daytime
Hypersomnolence-narcolepsy
Hypertension
Hypnagogic hallucinations
Retrograde amnesia
Personality change
Diminished vigor
Interpersonal/marital/job-related difficulties
Depression
? Females with obvious micrognathia
Enuresis
Family history of sleep-related breathing disorders
Insomnia
Allergic nasal obstruction
Morning headaches/confusion
Intellectual deterioration
Abnormality of sexual functioning, impotence, ↓ libido
History of facial or airway surgery
Excessive appetite

Airway Examination

Observation of the dynamic changes in the airway is desirable in patients with apnea secondary to obstruction of the airway. Guilleminault et al.⁴² in 1978 studied sleep-apneic adults during wakefulness and sleep by simultaneous electromyography and fiberoptic endoscopic examination of the nasopharynx. They demonstrated a striking posterior and lateral pharyngeal wall collapse associated with decreased electromyographic activity during obstructive events among these patients. They observed no increase in the number of episodes of apnea compared with noninvasive polysomnographic monitoring of the same patient on the previous night.

These observations are interesting and informative, but will probably be most useful at the research level. In the management of sleep-induced apneas, laryngoscopy is obviously of very limited usefulness.

Pneumography

The incidence of apnea in infancy is unknown. In 1975 Stein and Shannon⁴² reported the use of the pediatric pneumogram as an improved technique in investigating infantile apnea. The impedance pneumogram monitors up to 24 hours of cardiac and respiratory activity in a relatively noninvasive manner applicable to the hospital or home setting. Cassette-like tapes are subsequently reviewed. Based upon dual studies using simultaneous pneumography and alarm devices, alarm devices detected only about 40 per cent of the episodes of apnea.

Apnea is frequent among premature infants, as is the occurrence of the sudden infant death syndrome. The current working definition of the sudden infant death syndrome is: "the sudden death of any infant or young child, which is unexpected by history, and in which a thorough postmortem examination fails to demonstrate an adequate cause for death." The incidence of sudden infant death syndrome is 2 to 3 per 1000 live births, and a male predominance is unquestionable. Ninety-one per cent of these deaths occur before the infants are six months of age.³⁸

There has been an increasing interest in the association between apnea during sleep and the sudden infant death syndrome. There is evidence to support the association: Parents and caretakers of young infants dying of the sudden infant death syndrome report discovering them dead after or during periods when they were known or ex-

pected to be sleeping.⁴³ The child resuscitated from a "near miss" event has often been sleeping, is found apneic and blue, and with various degrees of intervention may start breathing again and recover.^{16, 44} Pulmonary petechial hemorrhages are characteristically found at postmortem examination in infants who have died of the sudden infant death syndrome. It is believed that they suggest a markedly negative intrathoracic pressure, as would be caused by trying to breathe against an obstruction. Also found at autopsy has been hypertrophy of pulmonary vasculature, which may suggest chronic hypoxia secondary to chronic hypoventilation prior to death.⁴⁵

Active sleep in infants (as in adults) is characterized by asynchrony of rib and abdominal respiratory movements. This effect, along with the low proportion of high-oxidative-capacity muscle fibers in the newborn diaphragm demonstrated by Keens et al.⁴⁶ and consequent fatigue, may contribute to the high susceptibility of infants to sleep-breathing difficulties.

Reports of comparisons of "near miss" infants, premature periodic breathers, and control infants have suggested that the first two groups have markedly increased apnea index values, respiratory pauses, and decreased carbon dioxide responsiveness. Most of these abnormalities disappear by the age of six months.¹⁷

A less clear-cut difference was found by Guilleminault et al.¹⁸ in a study in which 29 "near miss" full term infants were compared with controls. A statistically significant difference between "near miss" and control groups was found for only one parameter, the mixed-plus-obstructive events ($p < 0.001$).

Hoppenbrouwers and coworkers⁴⁷ confirmed the absence of any ≥ 15 second event among control infants one to six months of age and only a very rare occurrence from birth to one month of age. One of the difficulties in comparing such studies arises from the fact that several different criteria are used to define apnea. Two, 3, 5, 6, 10, 15, and 20 seconds are to be found as criteria in the literature.

The study of normal infants in natural environments has been limited.⁴⁶⁻⁴⁹ There is considerable information to suggest that the alteration of the environment by study in hospital sleep laboratories may alter the results obtained in sleep studies. The technique of pneumography has thus far found little use after infancy. This is partly because adults tolerate sleep studies adequately in a hospital setting. Motivation may play an important part in this tolerance. In

addition, although sleep disorders in adults may be life threatening in the long run, the immediate risk is low compared with the risk of the sudden infant death syndrome. Adaptations of these techniques may offer a great deal in the study of sleep disorders in the older child and adult.

Polysomnography

The techniques of polysomnography have been detailed. They are rapidly becoming the major diagnostic aid in the sleep apnea syndromes. Over the last several years a number of centers around the country have developed laboratories specializing in the study of the physiology and the pathophysiology of sleep-related disorders. More recently, clear standards for such laboratories have been established, in Certification Standards and Guidelines for Sleep Disorders Centers.⁵⁰ In most laboratories every effort is made to monitor without airway violation, unnecessary discomfort, etc., to avoid introducing variables into the system. Even without such devices, sleep laboratory experiences often contribute to the abnormalities found. This limitation is inherent and somewhat unavoidable. Many laboratories require at least two separate evenings of monitoring for acclimatization, discarding the results of the first evening. Needless to say, all such sleep monitoring must be done with the subjects unsedated.

Chemoreceptor Studies

Belief that problems with sleep-related apnea are related, at least in part, to abnormalities or alterations of the respiratory control mechanisms has led to investigation of chemoreceptor responses.⁵¹⁻⁵⁵ As previously discussed, sleep, especially REM sleep, is associated with a number of specific alterations in the responses to changes in P_{aO_2} , P_{aCO_2} , and pH in normal subjects. Numerous reports suggest that patients with what appear to be sleep-breathing dysfunctions have abnormal responses to these stimuli.⁵¹⁻⁵⁶ The methods employ altering FI_{O_2} and FI_{CO_2} (and at times acid-base status) while monitoring alveolar ventilation, arousal responses, blood gas tensions, and various pneumographic or polysomnographic indices.

Echocardiography

The echocardiogram may show alterations in sleep apneic patients suggesting right-sided car-

diac hypertrophy or even cor pulmonale. This is useful in indirectly establishing the abnormality of arterial blood gas tensions, and, thus, the severity of obstruction or abnormal central control. Occasionally the diagnosis of sleep disorder is suggested by a consistent finding on echocardiograms obtained for another reason. This examination, if the results are initially abnormal, may be used to confirm return to normal following therapy.

Hemodynamic Studies

Considerable abnormal variations in systemic arterial pressure and pulmonary arterial pressure have been demonstrated in apneic patients aside from the autonomic changes characteristic of REM sleep.⁵⁷ Owing to the invasiveness and risks of hemodynamic procedures, they are not routinely used for evaluation of patients with apnea.

Cardiac Monitoring

Tilkian et al.⁵⁸ used Holter monitoring in adult patients with documented sleep-induced apnea. Arrhythmias (including sinus bradycardia, asystole, sinus arrhythmia, ventricular and sinus tachycardias and second-degree atrioventricular block) were found in 96 per cent. They postulated that with these arrhythmias a small, but finite, number of patients can be presumed to have cardiac arrest. The prevalences suggested by their study are impressive, and the use of Holter monitoring should be considered for selected apnea patients.

Audiovideo Monitoring

Tape recordings of sleep both in the home and in the hospital have, at times, been helpful in screening patients who may need more in-depth investigation. In the sleep laboratory they may provide a useful adjunct to polysomnography or pneumography.

TREATMENT

The management of patients with disorders of breathing during sleep is dependent upon exact classification of the cause of the disorder. Definition of the obstructive, central or mixed nature of the problem is essential to avoid haphazard care.

Obstructive Disorders

Obviously, knowledge of the site and nature of an obstructive process is essential in selecting the approach to management. For example, a tumor mass should be removed, whereas an area of dynamic airway collapse will often require different management. Obstruction secondary to tonsilloadenoidal hypertrophy may be manageable by tonsillectomy and adenoidectomy or, if recurrent, may necessitate tracheostomy. Nasopharyngeal obstruction has at times been managed with nasopharyngeal tube placement at bedtime. The surgical management of obstructive problems should be by the procedure that requires the least invasiveness or disfigurement to achieve the desired effect.

Central Disorders

Patients who have abnormalities of central respiratory driving mechanisms are a very inhomogeneous group. Management at present continues to be somewhat less than satisfying.

The management of infantile apnea with various amounts of methyl xanthines (caffeine or theophylline-aminophylline), with or without monitoring, is often successful.⁵⁹⁻⁶⁴ However, in adult patients a variety of treatments (progesterone,^{65, 66} airway management,⁶⁷ theophyllines,⁶⁸ oxygen,⁶⁹ diaphragm pacing,^{56, 70} tricyclic antidepressants,¹² acetazolamide⁷¹) have been employed, with equally varied results. Controlled investigations with careful observation for side effects are badly needed in this area.

Mixed Apnea Disorders

The care of patients with mixed disorders is empirical. Any one or a combination of the methods available for obstructive or central disorders may work. As with the purely obstructive and purely central forms, with the mixed disorders, into which category most sleep apneas of adult patients fall, the trials of management modalities have been many, but controlled trials few. It would seem that attempts to relieve apnea of the mixed type would require both airway and central directed management. However, some feel that by relieving the obstructive "lesion," regression of the (perhaps secondary) central changes may occur. Alternatively, the minor improvement afforded by a single type of thera-

TABLE 5. Management Modalities for Apnea Secondary to Airway Obstruction

DISORDER	MANAGEMENT
Orfacial anomaly	Specific reconstruction
Severe tonsilloadenoidal hypertrophy	Tonsilloadenoidectomy, recurrence; tracheostomy
Nasal polyps	Polypectomy
Dynamic airway collapse	Weight reduction Treatment of primary disease such as endocrine dysfunction (if present) Nasopharyngeal tube Cervical collar Tonsilloadenoidectomy Submucous resection Tracheostomy ⁷²
Compressive airway collapse	Resection or other treatment of mass

py may allow a critical change in chemoresponsiveness. Naturally the least risky and invasive regimen should be tried first.

Tables 5 and 6 summarize the treatment modalities currently available.

SUMMARY

The entities among the spectrum of sleep-related breathing disorders have been reviewed. The sleep apnea syndromes constitute a newly appreciated clinical problem. The protean and usually nonspecific nature of presenting complaints was pointed out, along with the fact that significant sleep apnea has been diagnosed in apparently normal subjects. These factors, as

TABLE 6. Management Modalities for Apnea of Central Origin

DISORDER	MANAGEMENT
Apnea of prematurity or infancy	Adequate monitoring and provision for intervention Assurance of homeostasis (temperature control, glucose, electrolytes, oxygenation) Caffeine, theophylline, belladonna, atropine Continuous positive airway pressure Ventilatory assist (rarely)
Diaphragmatic incompetency	Diaphragm pacing Chronic nighttime ventilation
Intoxication (central depression)	Avoidance of alcohol or implicated drugs
Undifferentiated (↓ chemoresponsiveness of unknown etiology)	Theophylline Diaphragm pacing Progesterone Acetazolamide Tricyclic antidepressants

well as the relative newness of our appreciation of the problem, work additively to produce an underestimation of prevalence. Although a small group of patients will seek treatment knowing what problem they have, after reading of it in lay press, this number will undoubtedly reach a plateau as the more obvious patients are identified.

A very striking male predominance is characteristic of all reported series of sleep-related breathing disorders. This fact and the fact that such disorders as snoring, coronary artery disease and hypertension are much more common in males and in sleep-apneic patients than in the general population may not be unrelated. This speculation seems further supported by the knowledge that myocardial infarctions frequently occur during sleep. The postulation that myocardial infarction results from episodes of desaturation during a particular stage of sleep in the vulnerable patient, if true, could be of practical (therapeutic) value.

The increasing number of associations of a variety of upper airway abnormalities being brought to light is very interesting. Review reveals that frequently these patients have experienced significant weight gains just before the diagnosis of sleep apnea is made. The question should be raised as to whether the subclinical apnea causes the weight gain or the weight gain becomes the trigger for major apnea to appear. At any rate, excessive appetite is frequently seen in severely sleep-apneic patients, although the cause is unknown. The possibility exists that children with disorders such as Down's syndrome, Prader-Willi syndrome, hypothyroidism, and other endocrinologic aberrations associated with obesity and sleep apnea also have poor intellectual performance partly as a result of chronic sleep apnea.

Concern in the approach to the young infant has been paradoxically short-term and related to the risk of sudden infant death syndrome. This is in particularly striking contrast to the usual pediatric approach with concern about the future. As our understanding grows there will undoubtedly be an extension of our interest in the behavior of these children beyond the usual period of vulnerability to sudden infant death syndrome.

As previously mentioned, our appreciation of the clinical spectrum of disorders of breathing during sleep is growing, while our understanding of the basic mechanisms capable of explaining them is still incomplete. This situation may be similar to many in medicine, where early

awareness of similarities and associations among disorders evolves to further understanding and appreciation of their heterogeneity. At present we are limited to the rather descriptive phase of our understanding. Entities such as sudden infant death syndrome, hypersomnolence, snoring, and the pickwickian syndrome do appear related to disorders of respiratory control during sleep. They may ultimately prove more diverse (or more similar) than we now appreciate.

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